Pancreatic Cysts

Fernando Calmet

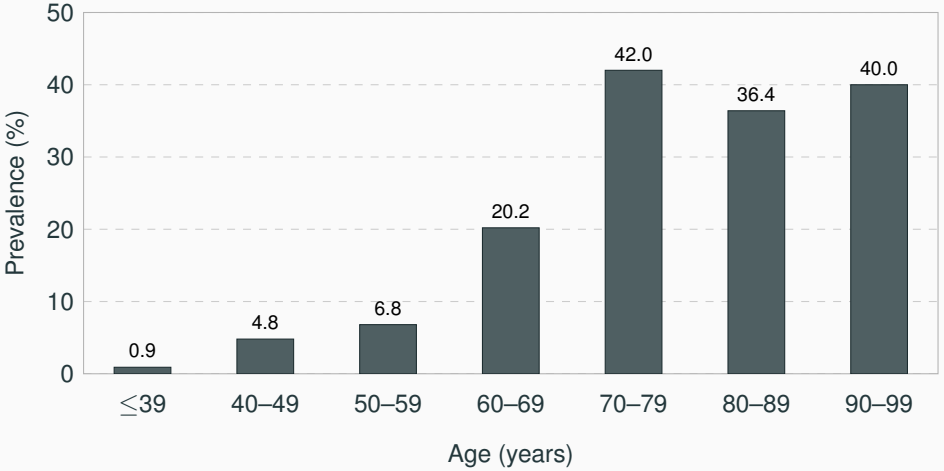
10/24/2026

Introduction

- Pancreatic cysts are common (6.3% of the population)
- Low risk of malignancy 0.24% per year (CI 0.12–0.36%)

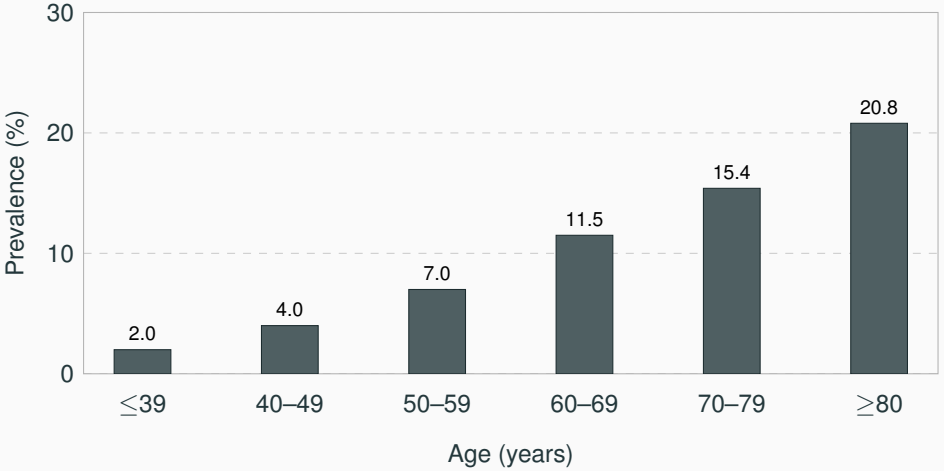
Wong et al. JAMA Netw Open. 2026;9(3):e260983.
Scheiman et al. Gastroenterology. 2015;148(4):824–48.

Prevalence of incidental pancreatic cysts by age group



Lee et al. Am J Gastroenterol. 2010;105(9):2079–84.

Prevalence of incidental pancreatic cysts by age group



Wong et al. JAMA Netw Open. 2026;9(3):e260983.

Classification

Premalignant

- Mucinous cystic neoplasm (MCN)
- Intraductal papillary mucinous neoplasm (IPMN)
- Intraductal oncocytic mucinous neoplasm (IOMN)
- Intraductal tubulopapillary neoplasm (ITPN)

Malignant

- Solid pseudopapillary neoplasm (SPN)
- Cystic degeneration of solid tumors

No malignant potential

- Serous cystadenoma (SCA)
- Pseudocyst
- Acinar cystic transformation
- Lymphoepithelial cyst
- Retention cyst

Gill et al. WHO classification of tumours: digestive system tumours. 5th ed. 2019. p. 295–372.

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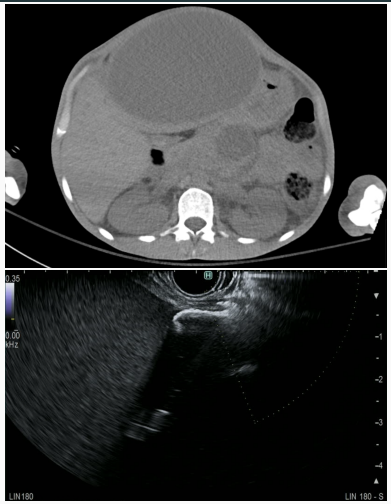
Pseudocyst

Overview

- More common in men (>75%)
- 4th and 5th decade of life
- Associated with antecedent acute pancreatitis
- No malignant potential

Management

- Pseudocysts do not require surveillance
- If symptomatic, may be treated by endoscopic ultrasound-guided cystogastrostomy



Ohtsuka et al. Pancreatology. 2024;24(2):255–70.
Sheth et al. Gastrointest Endosc. 2024;100(4):584–94.

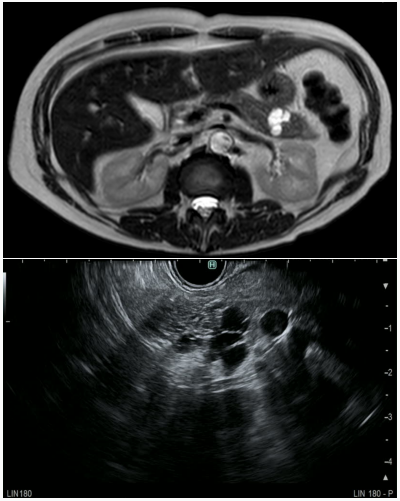
Serous cystadenoma (SCA)

Overview

- More common in women (70%)
- 6th and 7th decade of life
- May be macrocystic and/or microcystic
- No main pancreatic duct (MPD) communication
- Common in von Hippel-Lindau syndrome (VHL)
- Negligible malignant potential

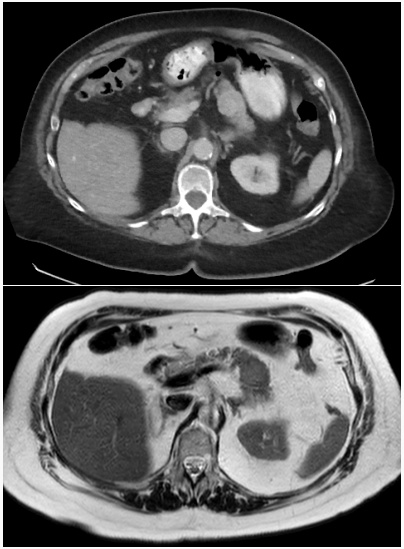
Management

- SCAs do not require surveillance



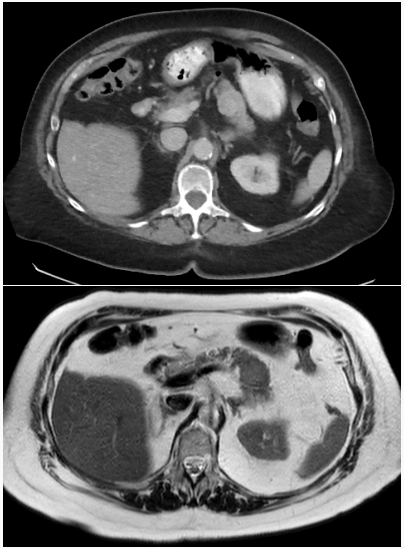
Ohtsuka et al. Pancreatology. 2024;24(2):255–70.
Lonser et al. Lancet. 2003;361(9374):2059–67.

Serous cystadenoma (SCA)





Serous cystadenoma (SCA)





Pathology: microcysts lined by small cuboidal cells with bland appearing nuclei and clear cytoplasm, consistent with a serous cystadenoma

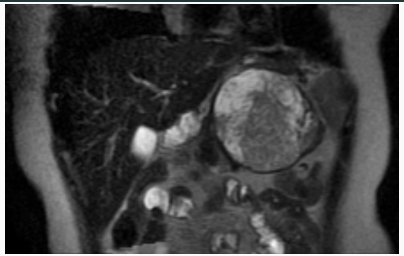
Solid pseudopapillary neoplasm (SPN)

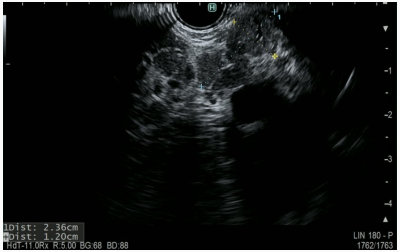
Overview

- More common in women (90%)
- 3rd and 4th decade of life
- Abdominal pain/discomfort common (63.8%)
- May have both solid and/or cystic components
- Low-grade malignancy
  - 7.7% metastases at presentation
  - 5-year survival: $\geq$98%

Management

- Resection $\rightarrow$ MRI yearly for at least 5 years





Elta et al. Am J Gastroenterol. 2018;113(4):464–79.
Law et al. Pancreas. 2014;43(3):331–37.

Mucinous cystic neoplasms (MCN)

Overview

- Almost exclusively in women ($>$95%)
- 4th and 5th decade of life
- Unilocular or multilocular
- Usually in the pancreatic body/tail (95%)
- Lack MPD communication

Management

- MCNs $<$4 cm without mural nodules
  $\rightarrow$ monitor vs. resection
- MCNs $\geq$4 cm or mural nodules $\rightarrow$ resection



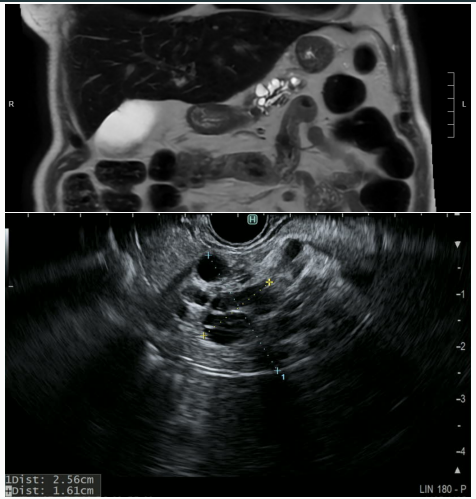


Ohtsuka et al. Pancreatology. 2024;24(2):255–70.
Elta et al. Am J Gastroenterol. 2018;113(4):464–79.

Intraductal papillary mucinous neoplasms (IPMN)

Overview

- More common in women (55%)
- 6th and 7th decade of life
- Involves main duct and/or branch duct
- MPD communication present, not always obvious

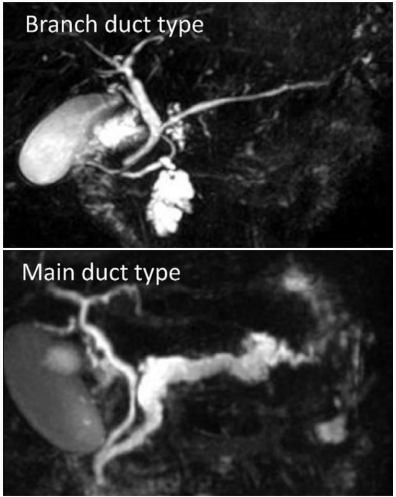


Ohtsuka et al. Pancreatology. 2024;24(2):255–70.
Elta et al. Am J Gastroenterol. 2018;113(4):464–79.

Intraductal papillary mucinous neoplasms (IPMN)

Classification of IPMNs

- Branch duct IPMN (BD-IPMN): pancreatic cysts with main duct communication
- Main duct IPMN (MD-IPMN): main pancreatic duct dilation $>$5 mm
- Mixed IPMN: pancreatic cysts with main pancreatic duct dilation



Gill et al. WHO classification of tumours: digestive system tumours. 5th ed. 2019. p. 295–372.
Ohtsuka et al. Pancreatology. 2024;24(2):255–70.

Risk of malignancy of IPMNs and MCNs

Surgical series

- BD-IPMN: 31% harbor HGD/invasive cancer (IC)
- MD-IPMN: 62% harbor HGD/IC
- MCN: 17% harbor HGD/IC
  - MCN $<$5 cm without mural nodules: 0% HGD/IC

Ohtsuka et al. Pancreatology. 2024;24(2):255–70.
Yun et al. Int J Surg. 2026;112(3):8214–27.
Marchegiani et al. Br J Surg. 2021;108(9):1097–104.

Risk of malignancy of IPMNs and MCNs

Surveillance studies

	Low-risk IPMNs*	Non-low-risk IPMNs†	MCNs
1 year	0.07%	1.95%	0.8%
3 years	1.73%	5.69%	
5 years	2.95%	9.77%	5.6%
10 years	7.11%	24.68%	36.5%

*IPMNs with MPD $<$5 mm and without mural nodule.

†IPMNs with MPD $\geq$5 mm and/or mural nodule

Choi et al. Clin Gastroenterol Hepatol. 2017;15(10):1509–1520.
Hozaka et al. Pancreas. 2023;52(5):e288–92.

Evaluation and surveillance of pancreatic cysts

General principles

- Pancreatic cysts are common
- Overall risk of malignancy is low
- Morbidity of pancreatic surgery is high
- Preoperative diagnosis of pancreatic cysts is suboptimal
- Surveillance should only be offered to patients who are fit for surgery

Work-up

Labs

- CA 19-9 and HbA1c levels: lack sensitivity/specificity

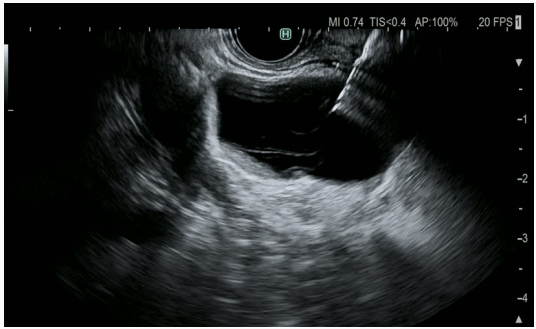
Imaging

- Contrast-enhanced MRI/MRCP
- Pancreatic protocol CT (if unable to undergo MRI)
- Endoscopic ultrasound (EUS) (in select circumstances)
    - Allows for cyst fluid sampling: glucose, CEA, amylase, next generation sequencing

EUS

Complications

- Perforation: 0.02–0.06%
- Bleeding: 0.13–0.69%
- Infection: 0.4–0.9%
- Pancreatitis: 0.32–0.92%
    - Severe: 0.04%



Forbes et al. Gastrointest Endosc. 2022;95(1):16–26.e2.
Wang et al. Gastrointest Endosc. 2011;73(2):283–90.

Pancreatic cyst fluid analysis—Non DNA-based markers

CEA

- High levels in mucinous cysts
- CEA $>$192 ng/mL: Sn 58%, Sp 87%

Glucose

- Low in mucinous cysts
- Glucose $<$50 mg/dL: Sn 93%, Sp 76%

Amylase

- Reflects communication with pancreatic duct
- Amylase $<$250 U/L: Sn 44%, Sp 98%

Pflüger et al. Pancreatology. 2023;23(7):868–877.
Scheiman et al. Gastroenterology. 2015;148(4):824–48.

Pancreatic cyst fluid analysis—DNA-based markers

Gene	Mucinous cyst		HGD/PDAC	
	Sensitivity	*Specificity*	*Sensitivity*	*Specificity*
KRAS	61%	99%	73%	35%
GNAS	44%	100%	49%	51%
KRAS + GNAS	79%	98%		
CDKN2A	6%	99%	11%	97%
PIK3CA	11%	100%	10%	97%
SMAD4	4%	98%	9%	98%
TP53	16%	99%	42%	95%

HGD, high-grade dysplasia; PDAC, pancreatic ductal adenocarcinoma.

Pflüger et al. Pancreatology. 2023;23(7):868–877.

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Gastroenterology 2015;148:819–822

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Santhi Swaroop Vege,[1] Barry Ziring,[2] Rajeev Jain,[3] Paul Moayyedi,[4] and the Clinical Guidelines Committee

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EC: Editor's Choice

SA-CME

Alec J. Megibow, MD, MPH[a], Mark E. Baker, MD[b], Desiree E. Morgan, MD[c], Ihab R. Kamel, MD, PhD[d], Dushyant V. Sahani, MD[e], Elliot Newman, MD[f], William R. Brugge, MD[g], Lincoln L. Berland, MD[c], Pari V. Pandharipande, MD, MPH[a,h]

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Pancreatology 24 (2024) 255–270

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Review Article

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Danish Pancreatic Cancer Group consensus guidelines on the management of pancreatic IPMN

Bahareh Abdolalizadeh[1], Claus Fristrup[2], Lars Peter Larsen[3], Eva Fallentin[1] and Caroline Ewertsen[1,4]

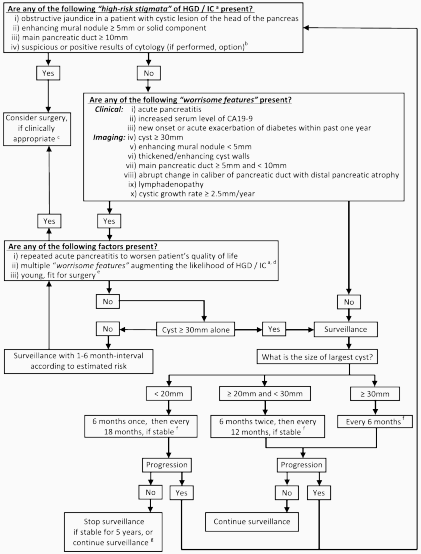
Guideline performance

		Sensitivity	*Specificity*	*PPV*	*NPV*
van Huijgevoort et al.	AGA 2015	26.9%	91.4%	63.6%	69.1%
n = 247	ESG 2018	96.2%	23.7%	41.32%	91.7%
	IAP 2024	94.2%	17.2%	38.9%	84.2%
Jung et al.	IAP 2017	100%	4.9%	42.4%	100%
n = 663	ESG 2018	97.8%	18.7%	45.7%	92.4%
	IAP 2024	95.6%	29.1%	48.6%	90.5%

AGA, American Gastroenterological Association; ESG, European Study Group on Cystic Tumours of the Pancreas; IAP, International Association of Pancreatology.

van Huijgevoort et al. Pancreatology. 2023;23(3):251–257.
Jung et al. Ann Surg. 2026;283(6):1044–1050.

IAP 2024 algorithm for the management of suspected BD-IPMN



Ohtsuka et al. Pancreatology. 2024;24(2):255–70.

Assessing risk of pancreatic cysts

High risk stigmata

- Obstructive jaundice in a patient with cystic lesion of the head of the pancreas
- Enhancing mural nodule $\geq$5 mm or solid component
- Main pancreatic duct $\geq$10 mm
- Suspicious or positive results of cytology (if performed)

Ohtsuka et al. Pancreatology. 2024;24(2):255–70.

Assessing risk of pancreatic cysts

Worrisome features

- Acute pancreatitis
- Increased CA 19–9
- New onset or worsening diabetes within past year
- Cyst $\geq$30 mm
- Enhancing mural nodule $<$5 mm
- Thickened/enhancing cyst walls
- Main pancreatic duct 5–9 mm
- Abrupt change in PD caliber with upstream pancreatic atrophy
- Lymphadenopathy
- Cyst growth $\geq$2.5 mm/year

Ohtsuka et al. Pancreatology. 2024;24(2):255–70.

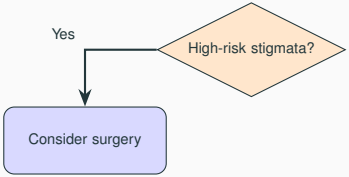
Surveillance of pancreatic cysts



High-risk stigmata
Obstructive jaundice with pancreatic head cyst
Enhancing mural nodule $\geq$5 mm
Main PD $\geq$10 mm
Suspicious/positive cytology

Adapted from Ohtsuka et al. Pancreatology. 2024;24(2):255–270.

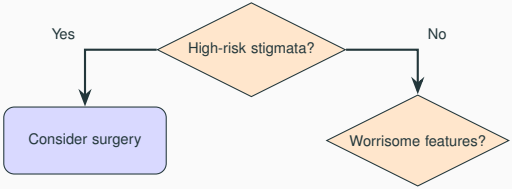
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Surveillance of pancreatic cysts



High-risk stigmata

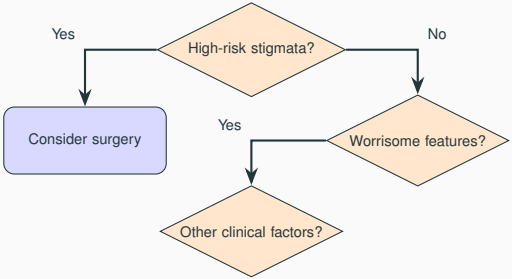
Obstructive jaundice with pancreatic head cyst
Enhancing mural nodule $\geq$5 mm
Main PD $\geq$10 mm
Suspicious/positive cytology

Worrisome features

Acute pancreatitis
Increased CA 19–9
New/worsening diabetes within past year
Cyst $\geq$30 mm
Enhancing mural nodule $<$5 mm
Thickened/enhancing cyst walls
Main pancreatic duct 5–9 mm
Abrupt change in PD caliber with upstream atrophy
Lymphadenopathy
Cyst growth $\geq$2.5 mm/year

Adapted from Ohtsuka et al. Pancreatology. 2024;24(2):255–270.

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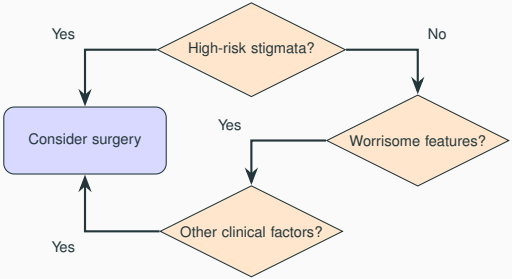
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Recurrent acute pancreatitis
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Young, fit for surgery

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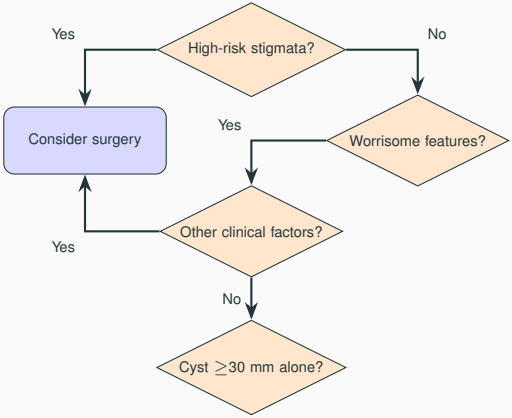
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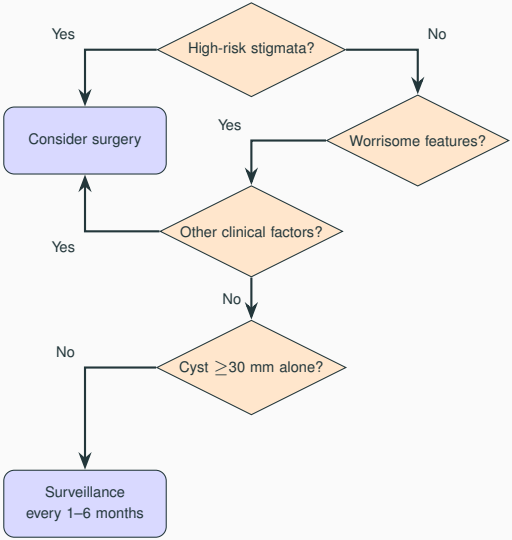
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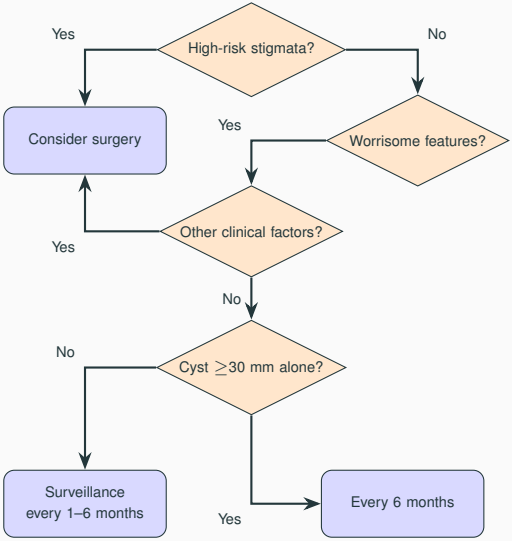
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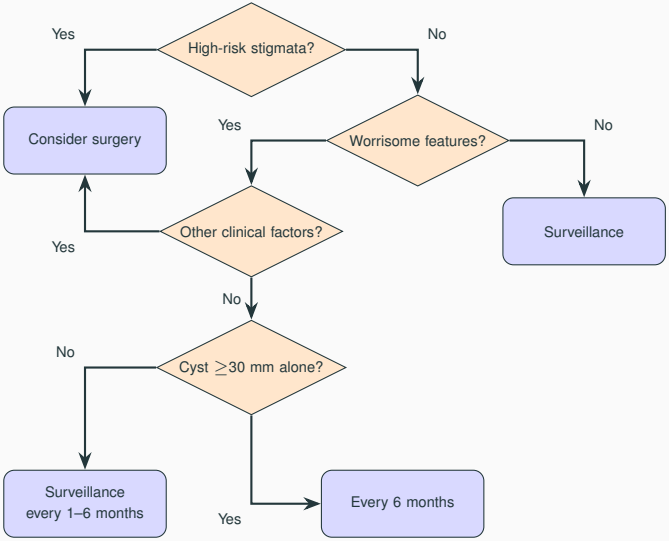
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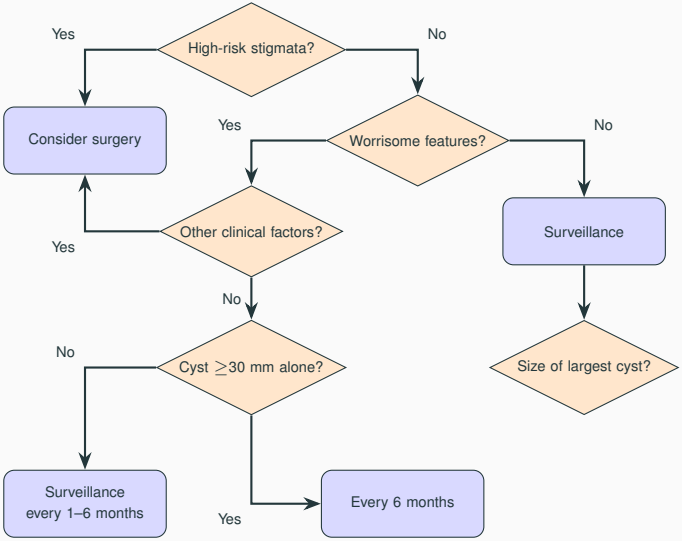
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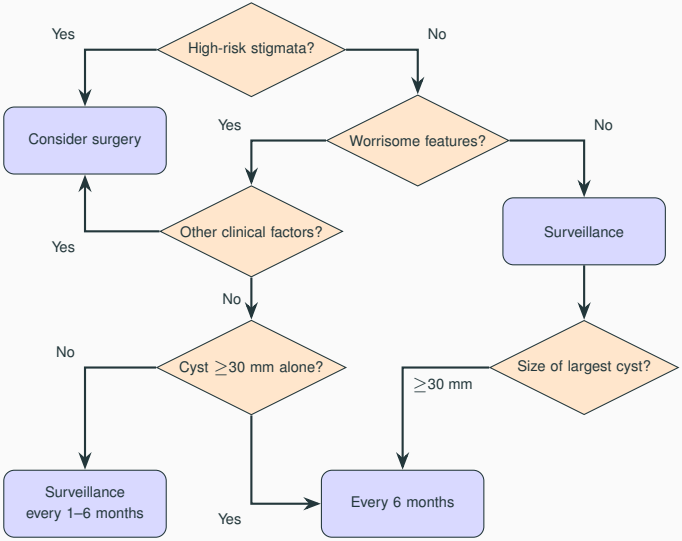
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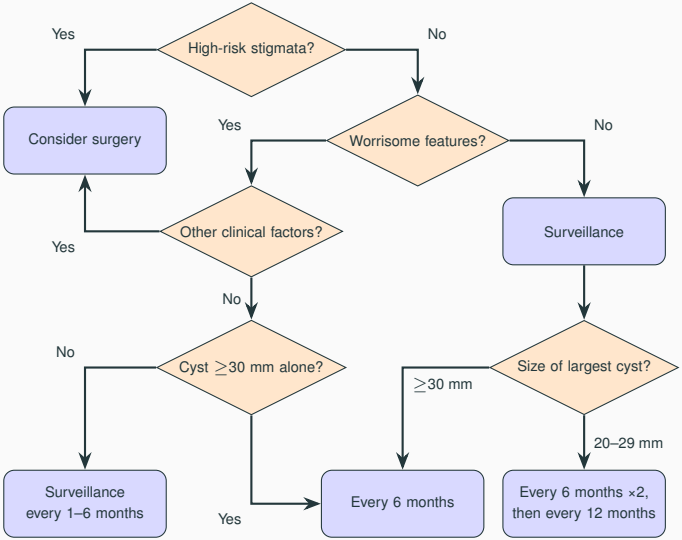
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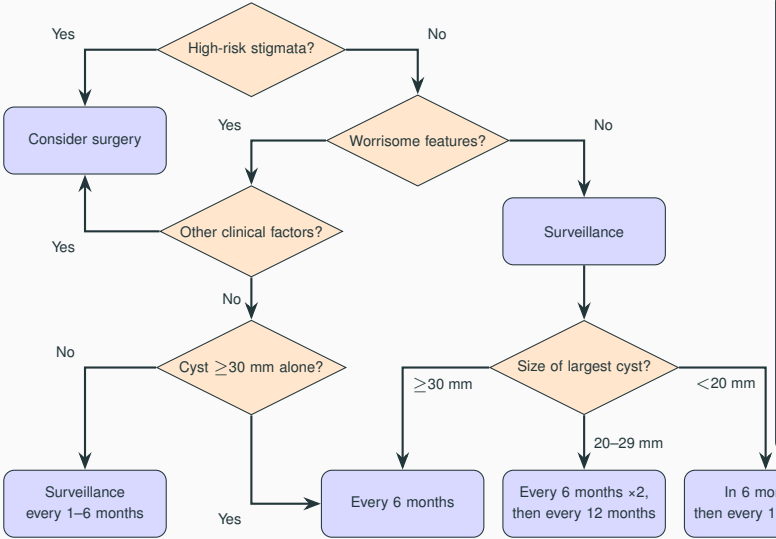
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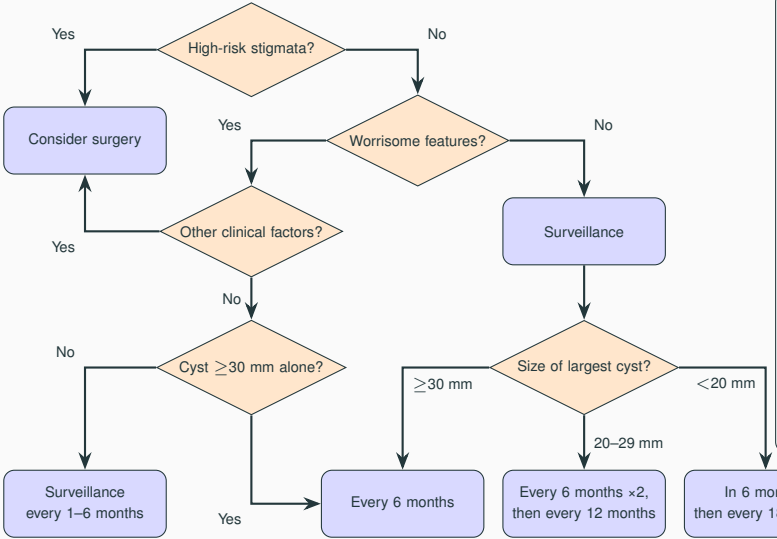
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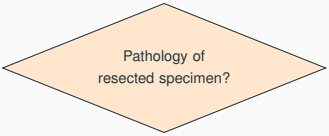
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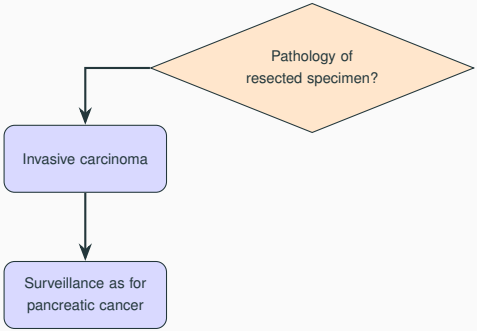
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Postoperative surveillance after IPMN resection



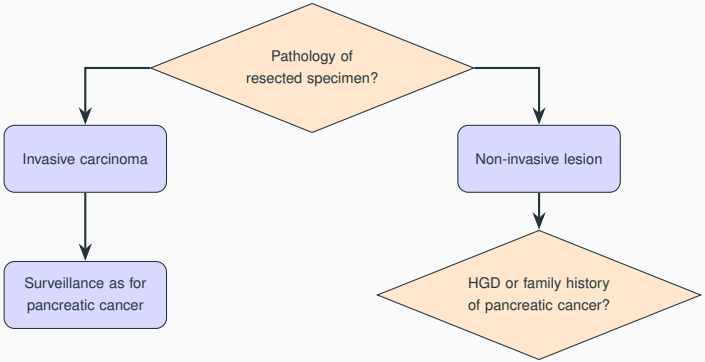
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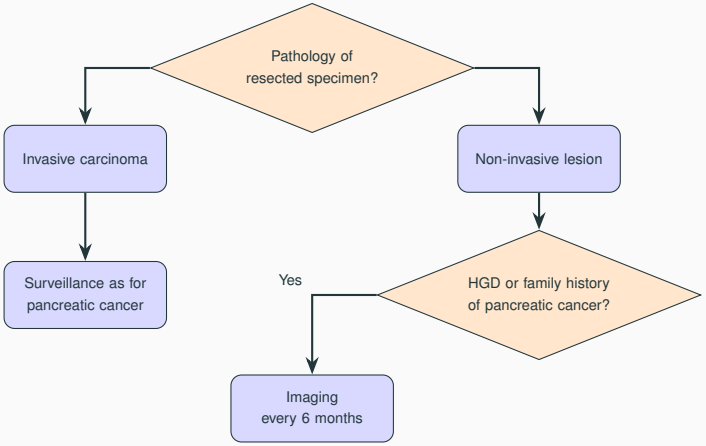
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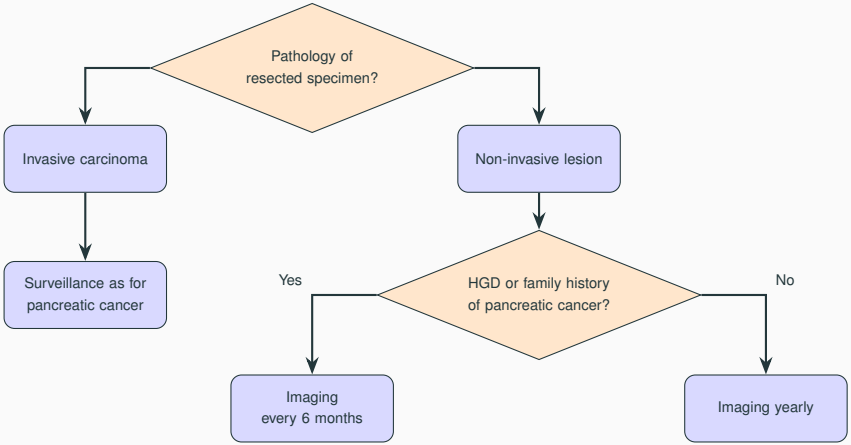
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Miscellaneous

Diagnostics

- Needle-based confocal laser endomicroscopy (nCLE)
- Through-the-needle biopsy
- Artificial intelligence

Jiang et al. Endosc Ultrasound. 2025;14(5):239–248.
Lee et al. Cancers (Basel). 2025;17(15):2558.
Cobb et al. In: Pancreatic cystic lesions: a multidisciplinary approach. 2026. p. 142–153.

Miscellaneous

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- Artificial intelligence

Cyst ablation

- Ethanol ablation
- Chemoablation: gemcitabine, paclitaxel
- Lauromacrogol ablation
- Radiofrequency ablation

Jiang et al. Endosc Ultrasound. 2025;14(5):239–248.
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